

where $h_{AB}(\Phi)$ is a new function of the Φ_n , but not of the Φ_n^* or derivatives.

The chiral gauge and scalar superfields are given by the expansions (26.3.21) and (27.3.15):

$$W_{AL}(x, \theta) = \lambda_{AL}(x_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L f_{A\mu\nu}(x_+) + (\theta_L^T \epsilon \theta_L) \not{D} \lambda_{AR}(x_+) - i \theta_L D_A(x_+),$$

$$\Phi_n(x, \theta) = \phi_n(x_+) - \sqrt{2} (\theta_L^T \epsilon \psi_{nL}(x_+)) + \mathcal{F}_n(x_+) (\theta_L^T \epsilon \theta_L),$$

where x_+^μ is the shifted coordinate (26.3.23). The terms of second order in θ_L (and independent of θ_R) in $\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL})$ are then

$$\begin{aligned} & - \left[\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\theta_L^2} = \\ & (\theta_L^T \epsilon \theta_L) \sum_{AB} (\lambda_{AL}^T \epsilon \lambda_{BL}) \left[\frac{1}{2} \sum_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} - \sum_n \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \right] \\ & + (\theta_L^T \epsilon \theta_L) \sum_{AB} h_{AB}(\phi) \left[- (\bar{\lambda}_A \not{D} (1 - \gamma_5) \lambda_B) - \frac{1}{2} f_{A\mu\nu} f_B^{\mu\nu} \right. \\ & \quad \left. + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + D_A D_B \right] \\ & + \sqrt{2} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} (\theta_L^T \epsilon \psi_{nL}) \left[- (\lambda_{BL}^T \epsilon \gamma^\mu \gamma^\nu \theta_L) f_{A\mu\nu} + 2i (\lambda_{BL}^T \epsilon \theta_L) \right], \end{aligned}$$

with all fields now understood to be evaluated at x^μ rather than x_+^μ . (The first and second terms on the right-hand side are taken from Eqs. (26.4.4) and (27.3.16), respectively.) Also, by writing $\theta_{L\alpha} \theta_{L\beta}$ as $\frac{1}{2} \epsilon_{\alpha\beta} (\theta_L^T \epsilon \theta_L)$, the third term on the right-hand side can also be expressed as proportional to $(\theta_L^T \epsilon \theta_L)$:

$$\begin{aligned} & (\theta_L^T \epsilon \psi_{nL}) \left[(\bar{\psi}_B \gamma^\mu \gamma^\nu \theta_L) f_{A\mu\nu} - 2i (\bar{\psi}_B \theta_L) \right] = \\ & \frac{1}{2} (\theta_L^T \epsilon \theta_L) \left[(\bar{\psi}_B \gamma^\mu \gamma^\nu \psi_{nL}) - 2i (\bar{\psi}_B \psi_{nL}) D_A \right]. \end{aligned}$$

The $\mathcal{F}$ -term is the coefficient of $(\theta_L^T \epsilon \theta_L)$, so

$$\begin{aligned} & - \left[\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\mathcal{F}} = \\ & \sum_{AB} (\lambda_{AL}^T \epsilon \lambda_{BL}) \left[\frac{1}{2} \sum_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} - \sum_n \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \right] \end{aligned}$$